

CSIT 345 – Operating Systems – HW2

Part A. Exercises (Textbook) – each 10 pts

4.16 A system with two dual-core processors has four processors available for scheduling. A CPU-intensive application is running on this system. All input is performed at program start-up, when a single file must be opened. Similarly, all output is performed just before the program terminates, when the program results must be written to a single file. Between start-up and termination, the program is entirely CPU-bound. Your task is to improve the performance of this application by multithreading it. The application runs on a system that uses the one-to-one threading model (each user thread maps to a kernel thread).

- How many threads will you create to perform the input and output? Explain.
- How many threads will you create for the CPU-intensive portion of the application? Explain.

4.17 Consider the following code segment:

```
pid_t pid;

pid = fork();
if (pid == 0) { /* child process */
    fork();
    thread_create( . . . );
}
fork();
```

- a. How many unique processes are created?
- b. How many unique threads are created?

**Part B. Programming Project – Scheduling Algorithms with the second Further Challenge (Chap 5) p. 29
– 80 pts**

You can use either C or Java to implement it.

Submission Guideline

1. Please submit a SINGLE PDF or Microsoft Word document.
2. Ensure your name is readable on the top of the first page.
3. It is fine that you write down your solutions on paper, take pictures, and include them in the document.
4. For each problem:
 - a. Identify clearly each problem through its number.
 - b. Explain the details or describe the procedures that you use to work out the problems.
5. For programming project:
 - a. Include source code files and the screenshots of results in the document.
6. Late submission – A late penalty of 10% of the total score for each day applies.